CER Prep Guide

How to prepare for, organize, and answer photosynthesis CER questions

Claim · Evidence · Reasoning — step by step, with model answers built from your own lab data

Your test will ask you to write **CER answers** about photosynthesis and cellular respiration. This guide shows you how to do that every time: a 5-step method, claim templates, a toolkit of science principles for your reasoning, three fully worked examples, and a test-day plan. Every example comes from your own packets, so the numbers will look familiar.

Packet codes: **[P1]** "Photosynthesis and Respiration" worksheet (Model 1, Q1–12; Model 2 carbon cycle, Q13–22) · **[P2]** "Photosynthesis & Cellular Respiration Simulation" lab, Parts I–VII (your team did Part III, Light Intensity) · **[P3]** your notes "Mauna Loa & the Rise of CO_2".

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1 · What CER is — and why teachers use it

CER stands for **Claim, Evidence, Reasoning**. It is how scientists argue: they give an answer, prove it with data, and explain the science that connects the two. A CER answer is not a summary of the whole lab. It is a short, focused argument that answers **one** question.

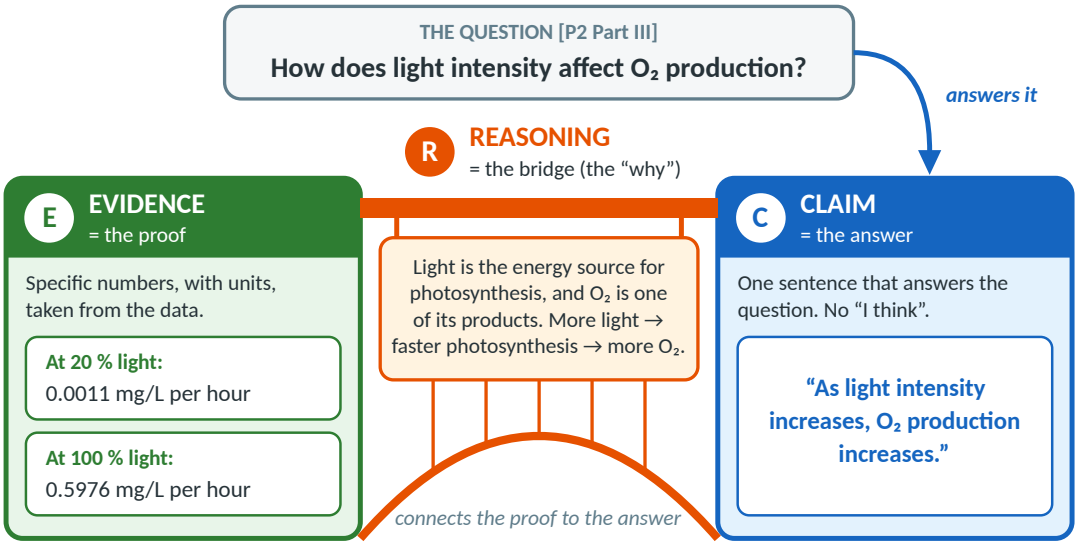


Figure 1. Anatomy of a CER answer. The claim answers the question, the evidence proves it, and the reasoning is the bridge that explains why the evidence supports the claim (example: [P2 Part III]).

Why teachers use CER. It shows real understanding: anyone can copy a number, but explaining *why* it came out that way proves you know the science. It is fair to grade, because each part earns its own points (Section 2). And it is this unit's big task: [P2 Part VII] asks you to "write a claim, provide evidence (numbers), and a scientific explanation incorporating photosynthesis and respiration. (Color code or label your C, E, and R.)"

> **→ TIP — Color code AND label**
>
> Claim = **blue [C]** · Evidence = **green [E]** · Reasoning = **orange [R]**. Always write the letters too, so your answer still works in pencil, on a black-and-white copy, and for color-blind readers. Every model answer in this guide is labeled this way.

2 · How CER answers are graded

Most 9th-grade CER rubrics look like this one (6 points). Your teacher's may differ slightly, but the habits that earn points are the same. Notice that **half the points are for reasoning.**

Part	0 points	1 point	2 points	3 points
Claim (0–1)	Missing, wrong, or does not answer the question	Accurate, complete, answers the question	—	—
Evidence (0–2)	None, wrong, or not related to the claim	Vague, or only one data point	At least 2 specific, accurate data points **with units** that support the claim	—
Reasoning (0–3)	Wrong science, or missing	Restates the evidence or is vague	Correct principle, but a weak link to the evidence	Correct principle(s) + **explicitly links** evidence to claim + correct vocabulary

Same question, two answers: 6/6 vs 3/6

Question: In Part III, why did the dissolved oxygen (DO) go *down* at 20 % light but go *up* at 100 % light? [P2 Part III]

Part	✓ 6/6 answer	× 3/6 answer
Claim	[C] At 20 % light the plant's respiration used more O_2 than photosynthesis made, so the DO fell; at 100 % light photosynthesis made more O_2 than respiration used, so the DO rose. **1/1** answers both halves of the question.	[C] The DO went down at 20 % because there wasn't much light, and it went up at 100 % because there was more light. **1/1** answers the question in one sentence.
Evidence	[E] At 20 % the DO fell by 0.1928 mg/L per hour, almost the same as in the dark run (−0.1939 mg/L per hour, Part II). At 100 % it rose by 0.4037 mg/L per hour (7.0064 → 7.4101 mg/L). **2/2** specific numbers with units, plus the dark-run baseline.	[E] At 100 % it went up by 0.4037. **1/2** only one number, with no unit (0.4037 *what*?). Nothing at all for 20 %.

Part	√ 6/6 answer	× 3/6 answer
Reasoning	[R] Plants do both processes at once. Respiration in the mitochondria uses O_2 all the time; photosynthesis in the chloroplasts uses light energy and releases O_2 as a product. The DO difference is the net: O_2 made − O_2 used. At 20 % photosynthesis added almost nothing (0.0011 mg/L per hour), so the DO fell as in the dark; at 100 % it made 0.5976, more than the 0.1939 used, so the water gained O_2. **3/3** both processes, organelles, O_2 as a product, and "net = made − used" tied to the numbers.	[R] Plants need light to do photosynthesis, which makes oxygen. **1/3** vague. Photosynthesis making O_2 cannot explain why the DO went *down*: respiration is missing, and nothing links back to the numbers.

To turn the 3/6 into a 6/6: add the 20 % numbers with units, name respiration (uses O_2 all the time, in the mitochondria), and explain "DO difference = O_2 made − O_2 used".

Score your own CER (use this on every practice answer)

	Check	Points
☐	**Claim** — one sentence, accurate, answers the exact question (no "I think")	1
☐	**Evidence** — at least one specific, accurate number **with its unit**	1
☐	**Evidence** — a second data point or a comparison ("from __ to __") that supports the claim	1
☐	**Reasoning** — names the correct science principle (e.g., light is the energy source; respiration uses O_2)	1
☐	**Reasoning** — explicitly links the evidence to the claim ("This shows… because… therefore…")	1
☐	**Reasoning** — correct vocabulary: reactant/product, chloroplast/mitochondrion, light energy/ATP	1
	Total	**6**

3 · The 5-step CER method

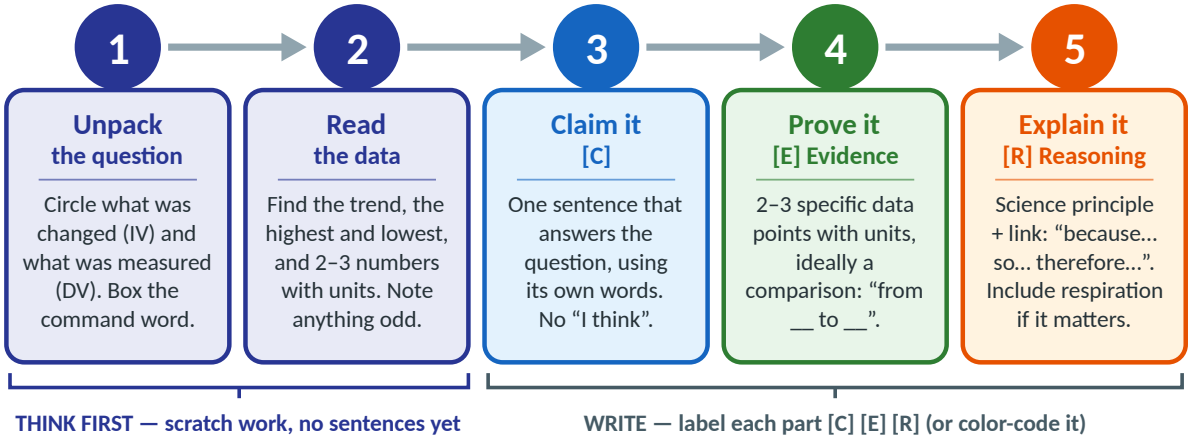


Figure 2. The 5-step CER method — use it on every CER question. Steps 1–2 are thinking (scratch work only); steps 3–5 are writing, one labeled part at a time.

Step	How to think: ask yourself...	Mini example (your Part III light data)	Watch out
1 · Unpack	What changed (IV)? What was measured (DV)? What does the command word (*describe*, *explain*, *compare*) want?	"How does **light intensity** (IV) affect **O_2 production** (DV)?" → a trend question → trend template.	Is it asking for O_2 production or DO difference?
2 · Read the data	What is the pattern? Highest? Lowest? Anything odd?	Flat near 0 at 20–40 %, then a steady rise; highest 0.5976 mg/L per hour at 100 %. Odd: 0.0011 (20 %) > 0.0002 (40 %), but both ≈ 0.	Per hour or per minute?
3 · Claim it [C]	Does my one sentence answer the exact question?	"As light intensity increases, O_2 production increases."	No "I think"; don't claim beyond the data.
4 · Prove it [E]	Could someone check these numbers in the table?	"O_2 production rose from 0.0011 mg/L per hour at 20 % to 0.5976 mg/L per hour at 100 %."	No unit = vague. Don't copy the whole table.
5 · Explain it [R]	WHY does the evidence support the claim? Which principle? Is respiration involved?	"Light is the energy source for photosynthesis: more light → more energy absorbed by chlorophyll → faster photosynthesis → more O_2 (a product)."	Don't just repeat the evidence.

4 · Writing strong claims

A strong claim answers the exact question, uses the question's own variables, gives a direction (increases/decreases), and stays inside what was tested. The question's wording tells you the template:

Question type	Claim template	Example from your packets
Trend — "How does X affect Y?"	"As [IV] increases, [DV] increases / decreases."	"As light intensity increases, O_2 production increases." [P2 Part III]
Best condition — "Which is best / optimal?"	"[Condition] gives the most [DV]."	"100 % light gave the most O_2 production of the light levels tested." [P2 Part III]
Comparison — "Compare A and B."	"[A] produced more [DV] than [B]."	"At 100 % light the plant produced more O_2 than at 60 % light." [P2 Part III]
Mechanism — "Why does … happen?"	"[Pattern] happens because [process]."	"CO_2 at Mauna Loa falls each summer because Northern-Hemisphere plants remove more CO_2 by photosynthesis than respiration and decay add." [P3]
Model-based — "What will happen if…?"	"[Action] will increase / decrease [quantity]."	"Burning fossil fuels will increase the amount of CO_2 in the atmosphere." [P1 Model 2, Q19]

Quick practice: write the claim (then check)

1. Which light intensity gave the most O_2 production in Part III? *(best condition)*

2. Why did the DO go down in the Part II dark run? *(mechanism)*

3. What would happen to atmospheric CO_2 if a large forest were cut down and burned? [P1 Q20] *(model-based)*

Answers

1. "100 % light gave the most O_2 production of the light levels tested (20–100 %)." Not "100 % is the best light for plants": you only tested up to 100 %.
2. "The DO went down in the dark because the plant's cellular respiration used O_2 while no photosynthesis was happening."

3. "Cutting down and burning a large forest would increase the amount of CO_2 in the atmosphere." (Fewer plants remove CO_2, and burning releases the trees' stored carbon.)

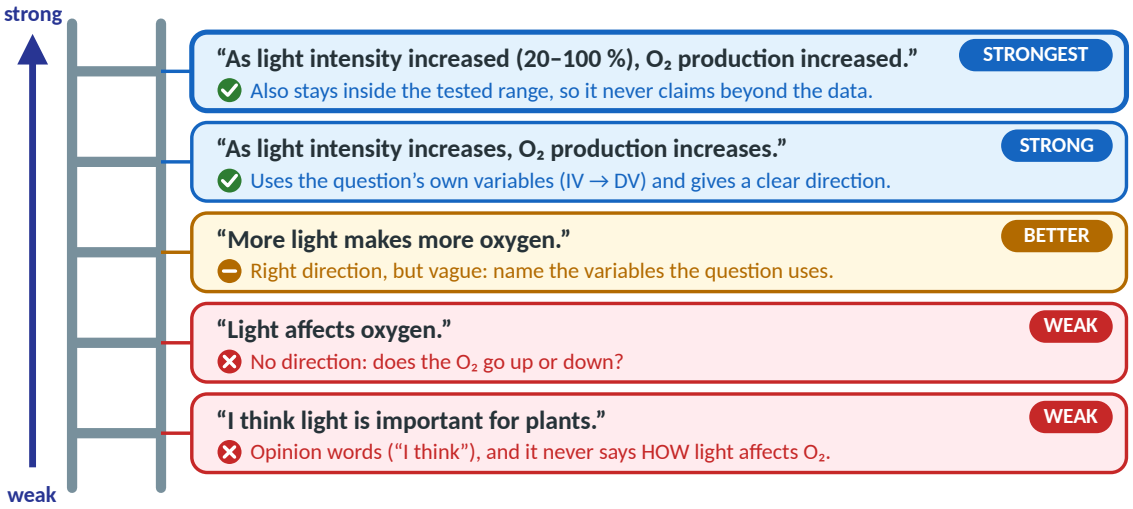


Figure 3. The claim ladder: climb from a weak claim to a strong one (your Part III question). Your own Q8 conclusion, "Higher light intensity generally means higher oxygen production", is already on rung 4 — nice! Section 9 turns it into a full CER.

5 · Choosing evidence

Evidence is the specific data that proves your claim. Choose the *right* numbers, not all of them.

How much evidence is enough?

- **2–3 data points, with units,** that directly support the claim. A change ("from 0.0011 to 0.5976 mg/L per hour") beats five numbers with no comparison. Describe the pattern in words too: "From 40 % up, each extra 20 % of light added about 0.20 mg/L per hour."
- **Include the baseline** when you explain a change: the Part II dark run (−0.1939 mg/L per hour) shows what respiration alone does.

Data quality: how far can you trust the numbers?

- **Trials and noise:** Part II averaged 3 trials (−0.1941, −0.1908, −0.1973 → −0.1939 mg/L per hour), which differ by up to 0.0065, so values smaller than about ±0.005 (like 0.0011 and 0.0002) can't be told apart from zero. Part III used only 1 trial per light level — a limitation worth naming.
- **Controls:** the Part II dark run is the baseline (respiration alone), and everything except light was kept the same (0 fish, 1 plant, 25 °C, white light, 60 minutes), so light caused the change in O_2 production.

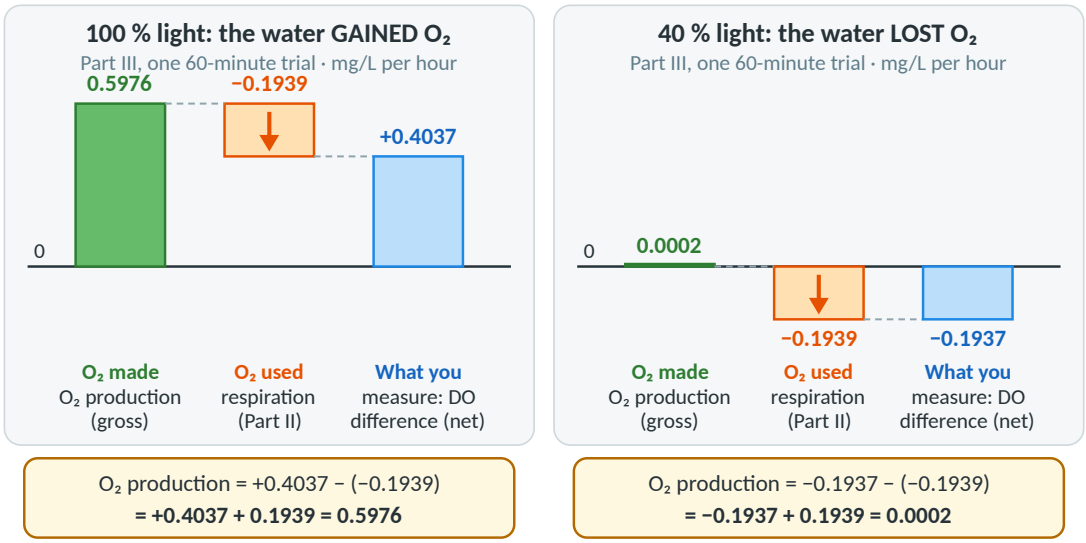
DO difference vs O_2 production — and the sign trap

Your table has two different "oxygen" numbers. The **DO difference** (= net change, end − start) is what you measured: O_2 made − O_2 used. **O_2 production** (= gross) is the total O_2 made by photosynthesis. The worksheet formula is *O_2 production = DO difference − respiration*. Because respiration is a negative number (−0.1939), subtracting it means **adding** 0.1939: −(−0.1939) = +0.1939. **Quote the right one, and name it:** how much O_2 the plant *made* → O_2 production (0.5976 at 100 %); what happened to the O_2 *in the water* → DO difference (+0.4037 at 100 %).

1	2	2	3	4	5
Light (%)	**DO start** (mg/L)	**DO end** (mg/L)	**DO difference** = end − start (mg/L per hour)	**O_2 production** = DO diff + 0.1939 (mg/L per hour)	**O_2 production** = per hour ÷ 60 (mg/L per minute)
100	7.0064	7.4101	+0.4037	0.5976 highest	0.0100
80	7.038	7.2464	+0.2084	0.4023	0.0067
60	7.0587	7.0633	+0.0046 ≈ 0	0.1985	0.0033
40	7.0009	6.8072	−0.1937	0.0002 ≈ 0	0.000003
20	7.0677	6.8749	−0.1928	0.0011 ≈ 0	0.00002

1. Independent variable (IV): what was changed. It starts your claim and goes on the x-axis.
2. Raw readings (mg/L). You use them to calculate; on their own they are rarely your best evidence.
3. DO difference = the NET change = O_2 made − O_2 used. "+" = the water gained O_2; "−" = it lost O_2. 60 %: +0.0046 ≈ 0, a break-even point.
4. O_2 production = the TOTAL O_2 made (gross). = DO difference − (−0.1939) = DO difference + 0.1939. Highest at 100 % (0.5976); ≈ 0 at 20 % and 40 %.
5. Per minute = per hour ÷ 60 (0.5976 ÷ 60 ≈ 0.0100). Use the time basis the question asks for. Always write which one you used.

Figure 4. How to read your Part III data table: five columns, five jobs.

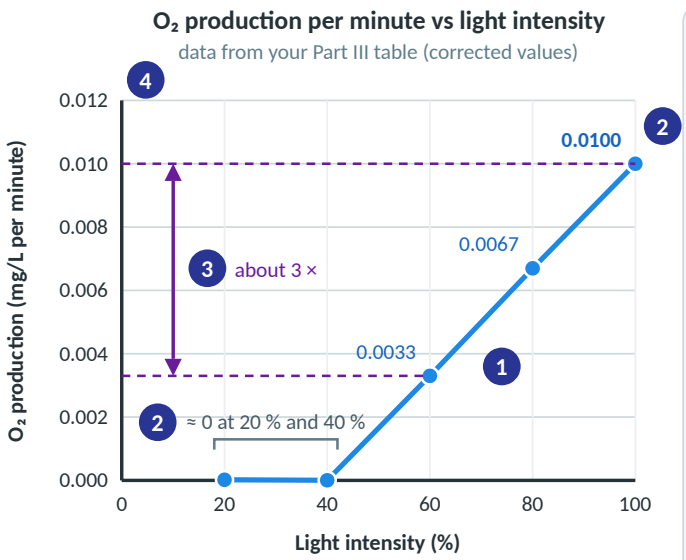


O_2 production = +0.4037 − (−0.1939)
= +0.4037 + 0.1939 = 0.5976

O_2 production = −0.1937 − (−0.1939)
= −0.1937 + 0.1939 = 0.0002

Minus a negative is a plus: O_2 production = DO difference + 0.1939 (mg/L per hour).

Figure 5. What you measure (DO difference) = O_2 made − O_2 used. Add back the respiration to get O_2 production. Data: your Part III table and Part II respiration rate.



FIVE MOVES FOR READING A GRAPH

1. **Trend**
   Describe the shape: flat near 0 from 20–40 %, then a steady straight-line rise.
2. **Highest and lowest**
   Highest: 0.0100 at 100 %.
   Lowest: ≈ 0 at 20 % and 40 %.
3. **Compare two points**
   60 % → 100 %: 0.0033 → 0.0100 mg/L per minute (about 3 times as much).
4. **Units**
   Read both axis titles: light in %; O_2 production in mg/L per minute.
5. **Time basis**
   This graph is per MINUTE. × 60 gives per hour: 0.0100 × 60 ≈ 0.60 (table: 0.5976).

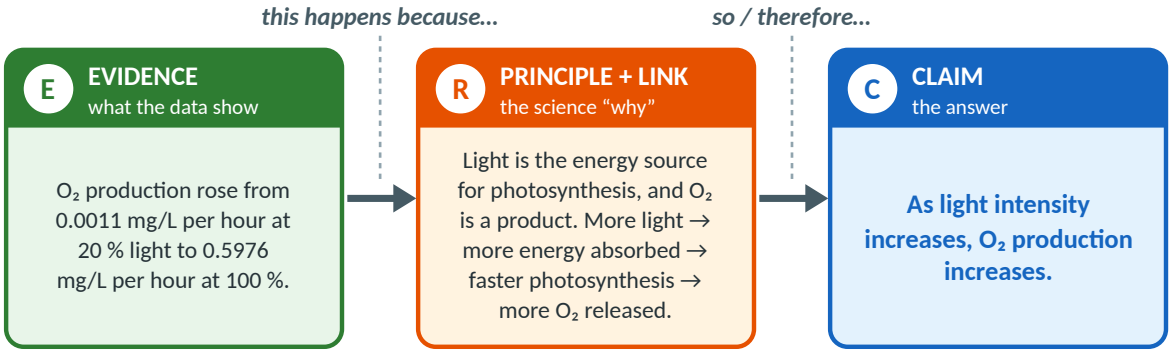
Figure 6. Five moves for reading a graph, shown on the correct graph of your Part III data (per minute).

6 · Building reasoning

Reasoning is worth the most points (3 of 6). It has two parts: a **science principle** (a rule that is always true) and a **link** showing how it explains *your* evidence. Pick the 1–3 principles that fit from this toolkit.

#	The Reasoning Toolkit: principle in one line († = standard biology a little beyond your packets)	Use it when...
1	**Light is the energy source** for photosynthesis; its energy is stored as chemical energy in glucose.	light intensity or color; "where does the energy come from?"
2	**O_2 is a product of photosynthesis:** more photosynthesis → more O_2 released.	your evidence is DO or O_2 production
3	**CO_2 is a reactant of photosynthesis:** photosynthesis removes CO_2 from the air or water.	Keeling Curve; carbon cycle; plant-and-fish tanks
4	**Respiration uses O_2 and releases CO_2 all the time** (light or dark), in the mitochondria, making ATP.	the DO fell; the dark run; fish; CO_2 rising in fall
5	**Plants do both** (chloroplasts AND mitochondria). Animals only respire. Decomposers respire too.	"Do plants need mitochondria?" [P1 Q5–10]
6	**Net vs gross:** DO difference = O_2 made − O_2 used; O_2 production = total O_2 made.	why the DO rose or fell; finding O_2 production
7 †	**Chlorophyll absorbs mostly red and blue/violet light and reflects green.** White light contains all colors.	light color (Part IV)
8 †	**Enzymes have an optimum temperature:** slower below it, less efficient above it; respiration speeds up when warmer.	temperature (Part V); "why not hotter?"
9 †	**Limiting factors:** the rate is held back by whichever factor is in shortest supply (light, CO_2, temperature).	"Why did it keep rising?" "What else limits it?"
10	**Carbon cycle balance:** without humans, CO_2 taken in by photosynthesis ≈ CO_2 returned by respiration and decay.	Model 2 questions [P1 Q18]

#	The Reasoning Toolkit: principle in one line († = standard biology a little beyond your packets)	Use it when…
11	**Fossil fuels release stored carbon:** burning moves ancient carbon into the air as CO_2; deforestation removes CO_2 absorbers.	human impact; the long-term rise [P1 Q19–20]
12	**The CO_2 level is a running total** (like water in a bathtub): it rises whenever more is added than removed.	why CO_2 peaks in May; why CO_2 keeps rising [P3]



Read it aloud: "The data show [E]. This happens **because** [principle]. **Therefore**, [C]."

Figure 7. The because-chain: evidence → principle → claim. Every link has to be there.

◊ HOW TO THINK ABOUT IT

Linking words that make the link visible: this shows that… · this happens because… · since… · therefore… · as a result… · this means… · …which is why… · at the same time… (more in Section 11). **The delete test:** cover your reasoning. If your answer still says the same thing, the reasoning only repeated the evidence.

7 · Counterclaim and rebuttal (CERR)

When a question gives an opposing idea ("A student says…") or asks you to *argue* or *evaluate*, add a **counterclaim** (a reasonable opposing idea, such as a watch-out from Section 10) and a **rebuttal** (evidence + reasoning that show why it is weaker). A good rebuttal can lift a strong answer to an excellent one.

Sentence frames, in order: (1) state the counterclaim fairly: "Some might argue that…" (2) point to evidence against it: "However, the data show…" (3) explain the science: "This is because…" (4) return to your claim: "Therefore…"

▪ EXAMPLE — Worked rebuttal: "Plants don't need mitochondria because they make their own food."

[C] Plants need mitochondria, because plant cells carry out cellular respiration. Some might argue that plants don't need mitochondria because they make their own food. [E] However, in the Part II dark run (0 % light, 0 fish, 1 plant) the DO fell in all three trials: −0.1941, −0.1908 and −0.1973 mg/L per hour (average −0.1939). [R] With no light there was no photosynthesis, and with no fish the plant was the only organism in the tank. The process that uses up O_2 is cellular respiration, which happens in the mitochondria. Making food is not the same as using it: the plant makes glucose by photosynthesis, but its cells still break that glucose down in their mitochondria to release energy as ATP. Therefore, the counterclaim is wrong: plants need both chloroplasts and mitochondria [P1 Q6].

8 · Three fully worked examples (think-aloud)

Try each question yourself first (about 10 minutes), then open the model answer. Each example shows the thinking for every step, a 6/6 model answer and why it earns 6/6, and two tempting wrong versions.

Example A · Light intensity and oxygen production

Data: your Part III table (Figure 4, Figure 6) · [P2 Part III] · question type: trend

Question. How does light intensity affect oxygen production during photosynthesis? Use your Part III data and explain your answer in terms of photosynthesis and respiration.

Step	What you think (out loud)
1 · Unpack	IV = light intensity (%); DV = O_2 production (mg/L per hour). "How does X affect Y?" → trend claim. "Photosynthesis and respiration" → my reasoning needs BOTH.
2 · Read the data	O_2 production 0.0011, 0.0002, 0.1985, 0.4023, 0.5976 (20 → 100 %): flat near 0, then up about 0.20 per +20 %. DO difference −0.1928 at 20 % but +0.4037 at 100 %.
3 · Claim	Trend template, plus "in the range we tested" so I don't claim beyond the data.
4 · Evidence	Lowest vs highest (0.0011 → 0.5976), the steady step size, and the DO difference switching from − to +.
5 · Reasoning	Toolkit #1 (light = energy), #2 (O_2 = product), #4 (respiration all the time), #6 (net vs gross), #9 (limiting factor), each linked to a number.

Model answer — Example A (6/6)

CLAIM
answers the question

In the range we tested (20–100 %), as light intensity increases, the plant's O_2 production increases.

EVIDENCE
specific data / facts

- O_2 production rose from 0.0011 mg/L per hour at 20 % light to 0.5976 mg/L per hour at 100 % light.
- From 40 % up, each extra 20 % of light added about 0.20 mg/L per hour (0.0002 → 0.1985 → 0.4023 → 0.5976).
- The water went from losing O_2 (DO difference −0.1928 mg/L per hour at 20 %) to gaining it (+0.4037 at 100 %).

REASONING
the science "why"

Light is the energy source for photosynthesis ($6CO_2 + 6H_2O$ + light energy → $C_6H_{12}O_6 + 6O_2$). Chlorophyll in the chloroplasts absorbs light energy, so more light means faster photosynthesis and more O_2, a product, released. At the same time the plant's mitochondria use O_2 for cellular respiration, light or dark (−0.1939 mg/L per hour in the Part II dark run). At low light respiration used more O_2 than photosynthesis made, so the DO fell; at high light photosynthesis made more than respiration used, so the DO rose. Because O_2 production was still rising steadily at 100 %, light was the limiting factor over the whole range we tested.

Why it scores 6/6. C (1) exact question, a direction, inside the tested range. E (2) three specific comparisons with units. R (3) energy source, product, organelles, respiration at the same time, every number linked.

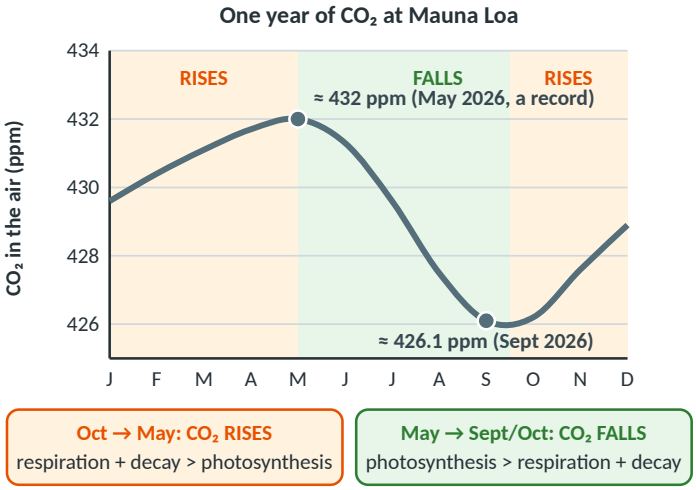
! WATCH OUT — Two tempting wrong versions

- *"Higher light means more oxygen because plants need light. At 100 % it was 0.5976."* → **3/6** (C 1 · E 1 · R 1). One number, no unit, no comparison. "Plants need light" has no energy, no product, no respiration.
- *"As light increases, O_2 production increases. At 100 % it was 0.4037 and at 40 % it was −0.1937, so at 40 % there was no photosynthesis. 200 % light would be even better."* → **1/6** (C 1 · E 0 · R 0). Quotes the DO difference as O_2 production (sign trap). The DO fell because respiration outpaced photosynthesis, not because it stopped. 200 % is beyond the data.

Example B · Why CO_2 peaks in May (the Keeling Curve)

Data: your notes [P3] + NOAA / Scripps monthly values · question type: mechanism

Question. The Keeling Curve shows that CO_2 at Mauna Loa rises and falls every year. It is highest in May and lowest in September–October. Explain this seasonal pattern using photosynthesis and cellular respiration.



Illustrative shape. Marked values: monthly averages, NOAA / Scripps (Sept 2026 preliminary). Yearly swing ≈ 6 ppm.

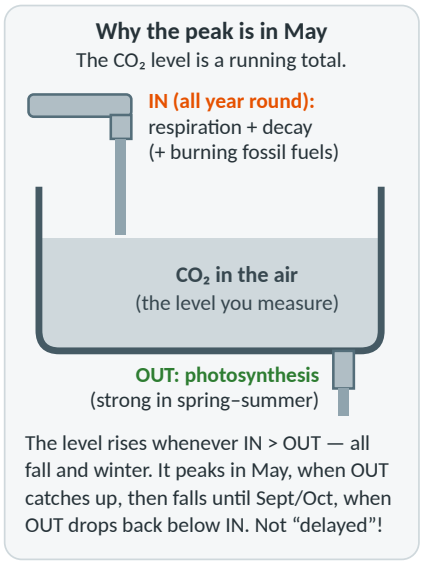


Figure 8. Left: one year of CO_2 at Mauna Loa (illustrative shape with two real monthly values). Right: the bathtub model — the CO_2 level is a running total.

Step	What you think (out loud)
1 · Unpack	"Explain this pattern" → mechanism claim: "[Pattern] happens because [process]." Use both processes. The "variable" is the season.
2 · Read the data	Peak ≈ 432 ppm (May 2026); low ≈ 426.1 ppm (Sept 2026, preliminary): about 6 ppm lower. Repeats every year; the Southern Hemisphere is opposite and weaker [P3].
3 · Claim	Mechanism template. Trap: my notes say "photosynthesis is a delayed reaction" (watch-out 8). The real reason is the running total.
4 · Evidence	The two monthly values in ppm, the size of the swing, and the timing (high in May, low in Sept/Oct).
5 · Reasoning	Toolkit #3 (CO_2 is a reactant), #4 (respiration and decay all year), #12 (running total). Explain the fall, the rise, and why the turn comes in May.

Model answer — Example B (6/6)	
CLAIM answers the question	CO_2 at Mauna Loa peaks in May and is lowest in September–October because Northern-Hemisphere plants remove CO_2 by photosynthesis mainly in spring and summer, while respiration and decay add CO_2 all year.
EVIDENCE specific data / facts	• CO_2 peaked at about 432 ppm in May 2026 (a record) and fell to about 426.1 ppm by September 2026 (NOAA, preliminary): a drop of about 6 ppm in one growing season. • The pattern repeats every year, and the Southern Hemisphere, with less land and fewer land plants, shows the opposite, weaker pattern [P3].
REASONING the science "why"	Photosynthesis uses CO_2 as a reactant ($6CO_2 + 6H_2O$ + light energy → $C_6H_{12}O_6 + 6O_2$). From May to September, trees in full leaf remove more CO_2 than respiration and decay add, so the level falls. In fall the leaves drop and photosynthesis slows, but cellular respiration by plants, animals and especially decomposers breaking down fallen leaves keeps releasing CO_2, so from October to May more CO_2 is added than removed. The CO_2 level is a running total, like water in a bathtub: it keeps rising while more flows in than out, peaks in May when photosynthesis catches up, and is lowest in Sept/Oct when photosynthesis drops back below respiration. Photosynthesis is not "delayed"; the level just takes months to build up and draw down.

Why it scores 6/6. **C (1)** pattern + cause. **E (2)** two real values with units and dates, the swing, the hemispheres. **R (3)** CO_2 as a reactant, respiration and decay all year, and the running total that explains the *timing* — the hardest part.

! WATCH OUT — Two tempting wrong versions

- *"CO_2 is higher in spring because photosynthesis is a delayed reaction: it takes time to remove CO_2."* → **0/6** (C 0 · E 0 · R 0). Keeps a mistaken note (watch-out 8): photosynthesis starts within seconds of light. No numbers, no respiration.
- *"CO_2 goes up and down because people burn more fossil fuels in winter. Respiration is constant, so it has a negligible effect."* → **0/6** (C 0 · E 0 · R 0). Fossil fuels explain the long-term rise, not the yearly zig-zag. Respiration plus decay release about as much CO_2 each year as photosynthesis removes (watch-out 8).

Example C · Optimal ecosystem productivity — and how many fish?

Data: class results [P2 Part VI] + your Part III data + HYPOTHETICAL PRACTICE DATA for the fish · [P2 Part VII] · question type: best condition + model

Question [P2 Part VII]**.** After your tests, what were the optimal conditions and why? How many fish could that environment support? Write a claim, provide evidence (numbers), and a scientific explanation incorporating photosynthesis and respiration. (Color code or label your C, E, and R.)

Step	What you think (out loud)
1 · Unpack	Two questions in one: (a) optimal conditions, (b) how many fish. "Photosynthesis and respiration" → both in my reasoning. "Label your C, E, and R" → write [C] [E] [R].
2 · Read the data	Class results: 100 % light, white light, 20 °C best tested (only 15, 20, 25, 30, 35 °C). One plant at 100 % white light: DO difference +0.4037 mg/L per hour = the O_2 "budget" for fish. Fish (HYPOTHETICAL): 0.10 each.

Step	What you think (out loud)
3 · Claim	Best-condition template + a number: 100 % light, white light, about 20 °C, about 4 fish. Say "best tested".
4 · Evidence	Light: 0.0011 → 0.5976 mg/L per hour. Color, temperature: class results (+ real Part IV/V numbers). Fish: 4 × 0.10 = 0.40 ≤ 0.4037; 5 × 0.10 = 0.50 > 0.4037.
5 · Reasoning	Toolkit #1, #7, #8 for the conditions; #4, #5 for the fish (fish only respire; healthy only while O_2 made ≥ O_2 used). Bonus: fish CO_2 feeds the plant [P1 Q10].

Which numbers are real? Your Part III data and the class results (100 % light, white light, about 20 °C) are real; get the missing Part IV/V numbers from classmates. The O_2 one fish uses (0.10 mg/L per hour) is **HYPOTHETICAL PRACTICE DATA**; your Part VII runs give the real value.

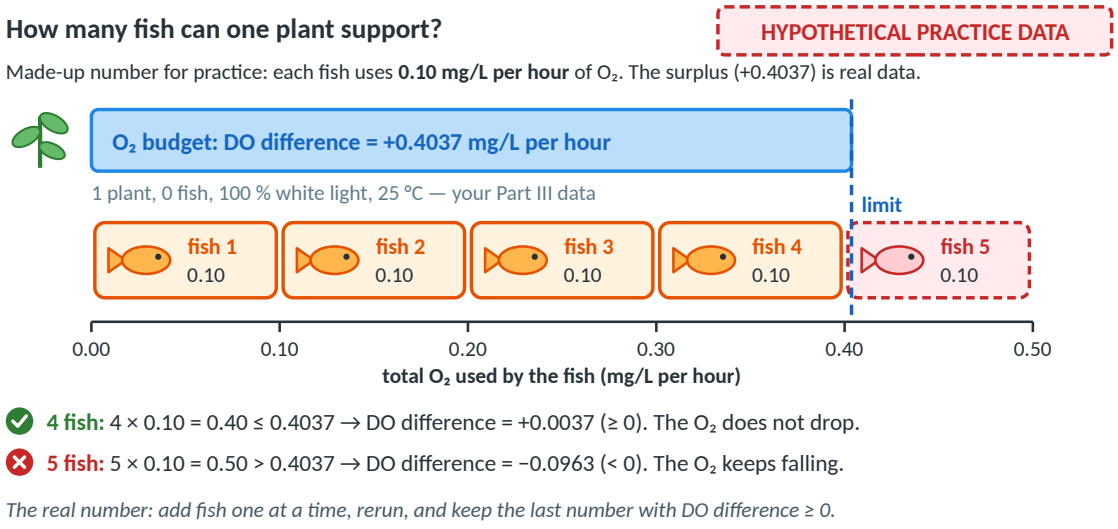


Figure 9. The fish "O_2 budget". The fish numbers are HYPOTHETICAL PRACTICE DATA; the +0.4037 surplus is your real Part III value (1 plant, 100 % white light, 25 °C).

Model answer — Example C (6/6), written as a labeled paragraph

> [C] The optimal conditions were 100 % light intensity, white light, and about 20 °C (the best temperature we tested), and under these conditions one plant could support about 4 fish. [E] In Part III, O_2 production rose from 0.0011 mg/L per hour at 20 % light to 0.5976 mg/L per hour at 100 %. The class results showed white light gave the most O_2 of the colors tested and 20 °C was best of the five temperatures tested (15–35 °C) [add your class's Part IV and V numbers here]. One plant at 100 % white light (25 °C, Part III) had a DO difference of +0.4037 mg/L per hour, an O_2 surplus. Using HYPOTHETICAL PRACTICE DATA of 0.10 mg/L per hour per fish, 4 fish use 0.40 (DO difference +0.0037, still ≥ 0), but 5 fish use 0.50 (DO difference −0.0963, so the O_2 falls). [R] Photosynthesis in the chloroplasts uses light energy, CO_2 and H_2O to make glucose and releases O_2 as a product. Bright light supplies the most energy, white light contains the red and blue light that chlorophyll absorbs best, and the enzymes of photosynthesis work best near their optimum temperature. Cellular respiration in the mitochondria of the plant and the fish uses O_2 all the time, and fish only respire. The tank stays healthy only while O_2 made ≥ O_2 used (DO difference ≥ 0), so the fish's total respiration must fit inside the plant's surplus of about 0.40 mg/L per hour: 4 fish fit, 5 do not. The fish also release CO_2, a reactant the plant needs, so plant and fish depend on each other.

Why it scores 6/6. **C (1)** both parts (conditions + fish). **E (2)** real numbers with units, class results, and labeled practice numbers. **R (3)** both processes, organelles, reactants/products, and the rule "O_2 made ≥ O_2 used" that turns numbers into a fish count.

→ REASONABLE INFERENCE (beyond the packets) — Level-4 thinking

The +0.4037 surplus was measured at 25 °C; at the best tested 20 °C it is probably a little larger, so rerun to check. At night there is no photosynthesis, so no surplus, and the DO falls. And 4 fish leave a margin of only 0.0037 mg/L per hour (about the size of the trial-to-trial spread), so a careful scientist might choose fewer fish.

! WATCH OUT — Two tempting wrong versions

- *"The best conditions are 100 % light, white light and 20 °C because plants need light. The tank could hold lots of fish."* → **1/6** (C 0 · E 0 · R 1). No numbers. "Lots of fish" does not answer "how many". No respiration, though the question asks for it.
- *"20 °C is best because above 20 °C the enzymes denature. More fish is better because fish make CO_2, so 10 fish would make the plant grow fastest."* → **0/6** (C 0 · E 0 · R 0). Enzymes don't denature just above 20 °C (watch-out 10). It ignores that fish use O_2: with the practice data, 10 fish would use 1.0 mg/L per hour, over twice the surplus.

9 · Upgrade a weak answer

Use the rubric like a repair manual: check the claim, then the evidence, then the reasoning, and fix the weakest part first. Try each upgrade before you open the answer.

Weak answer 1 — your own Part III conclusion [P2 Part III Q8]

"Higher light intensity generally means higher oxygen production." (About 1/6 as a CER: a good claim, but no evidence and no reasoning.)

- **Fix C:** keep it (it has a direction); say "O_2 production" and swap "generally" for the tested range.
- **Fix E:** add a comparison with units: 0.0011 → 0.5976 mg/L per hour; DO difference −0.1928 → +0.4037.
- **Fix R:** light = energy source; O_2 = product; respiration uses O_2 at the same time.

Upgraded answer 1 (6/6)

[C] In the range we tested (20–100 %), higher light intensity means higher O_2 production. **[E]** O_2 production rose from 0.0011 mg/L per hour at 20 % light to 0.5976 mg/L per hour at 100 %, and the DO difference changed from −0.1928 to +0.4037 mg/L per hour. **[R]** Light is the energy source for photosynthesis, so more light lets the chloroplasts make glucose faster and release more O_2, a product. The plant's mitochondria use O_2 for respiration all the time, so at low light the DO fell; at high light photosynthesis made more O_2 than respiration used, so the DO rose.

Weak answer 2 — the temperature reason [P2 Part VI]

"20 °C is best because at higher temperatures the enzymes are denatured." (About 1/6: a claim with no numbers and a science error.)

- **Fix C:** say "best *tested*": only 15–35 °C in 5 °C steps were tested, so the true optimum is between 15 and 25 °C.
- **Fix E:** get the class's Part V numbers: O_2 production at 20 °C vs a cooler and a warmer temperature, with units.
- **Fix R:** above the optimum, enzymes work less efficiently (real denaturing needs roughly 40 °C+), and respiration speeds up.

Upgraded answer 2 (6/6 once you fill in your class's numbers)

[C] Of the temperatures we tested (15–35 °C), 20 °C gave the most O_2 production. [E] At 20 °C, O_2 production was ___ mg/L per hour, compared with ___ mg/L per hour at 15 °C and ___ mg/L per hour at 30 °C [your class's Part V data]. [R] Photosynthesis is run by enzymes, which work fastest near an optimum temperature. Below it, molecules move and collide more slowly, so the reactions slow down. Above it, the enzymes work less efficiently (real denaturing needs roughly 40 °C or more), and the plant's respiration speeds up and uses more O_2, so the water gains less O_2.

Weak answer 3 — burning fossil fuels [P1 Model 2, Q19]

Question: How does burning fossil fuels affect the carbon cycle? **Weak answer:** "It makes pollution and is bad for the environment." (0/6: true-ish, but it does not answer the question.)

- **Fix C:** use the model-based template: "[Action] will increase / decrease [quantity]."
- **Fix E:** Model 2's combustion arrow + real data: about 313 ppm (1958) → about 432 ppm (May 2026); about 280 ppm before the Industrial Revolution.
- **Fix R:** natural balance [P1 Q18] + stored carbon released faster than photosynthesis removes it + running total.

Upgraded answer 3 (6/6)

[C] Burning fossil fuels increases the amount of CO_2 in the atmosphere and upsets the balance of the carbon cycle. [E] In Model 2, combustion moves carbon from carbon stores (coal, oil, natural gas) into atmospheric CO_2 [P1 Model 2]. At Mauna Loa, CO_2 rose from about 313 ppm in 1958 to about 432 ppm in May 2026, far above the pre-industrial level of about 280 ppm (Scripps / NOAA). [R] Without humans, the CO_2 taken in by photosynthesis roughly equals the CO_2 returned by respiration and decay, so the level stays about constant [P1 Q18]. Burning fossil fuels adds carbon stored for millions of years to the air as CO_2, faster than photosynthesis can remove it. Because the CO_2 level is a running total, adding more than is removed each year makes it keep rising (recently about 2–2.5 ppm per year).

10 · Common mistakes (and one-line fixes)

#	Watch-out	One-line fix
1	**Swapping the energy labels** (light in for photosynthesis, ATP out for respiration)	Photosynthesis takes **in** light energy; respiration **releases** energy as ATP.
2	**Forgetting plants also respire**	Plants have chloroplasts AND mitochondria; they respire 24/7, light or dark.
3	**Saying animals lack mitochondria**	Animals have mitochondria; what they lack is chloroplasts.
4	**Mixing up DO difference (net) and O_2 production (gross)** — and the minus-a-negative sign trap	O_2 production = DO difference − (−0.1939) = DO difference + 0.1939.
5	**Evidence without numbers/units**	Every evidence sentence gets a number + unit (mg/L per hour, %, °C, ppm).
6	**Reasoning that just repeats the evidence**	Add a principle ("because light is the energy source…") and link it.
7	**Claiming beyond the data** (e.g., "200 % light would be even better")	Say "in the range we tested (20–100 %)".
8	**Keeping the mistaken notes** ("respiration is negligible", "photosynthesis is delayed")	Respiration + decay release about as much CO_2 as photosynthesis removes; the level is a running total.
9	**"Respiration = breathing"**	Breathing moves gases; cellular respiration is a chemical reaction in mitochondria.

#	Watch-out	One-line fix
10	**Saying enzymes "denature" just above 20 °C**	Above the optimum they work less efficiently; denaturing needs roughly 40 °C+.
11	**Graph not matching the table / uneven scale / drawing beyond the data**	IV on x, DV on y, even ticks, mark every point, stop at the last point.
12	**Forgetting units and the time basis** (per hour vs per minute)	Write "mg/L per hour" or "mg/L per minute"; per minute = per hour ÷ 60.

11 · Sentence starters and transition words

Borrow these when you get stuck. They work for any CER question.

Part	Sentence starters	Linking / transition words
[C] Claim	"As ___ increases, ___ increases / decreases." · "___ gave the most ___." · "___ happens because ___." · "___ will increase / decrease ___."	none needed: just answer the question
[E] Evidence	"The data show that…" · "In Part ___, ___ rose from ___ to ___ (units)." · "At ___, ___ was ___, compared with ___ at ___." · "According to Model 2…"	for example · compared with · from … to … · while · in contrast
[R] Reasoning	"This happens because…" · "Since ___ is a reactant / product of ___, …" · "This shows that…" · "At the same time, respiration…"	because · since · therefore · so · as a result · this means · which is why
Counterclaim & rebuttal	"Some might argue that…" · "However, the data show…" · "This idea is weaker because…"	however · although · even though · on the other hand
Wrap-up (optional)	"Therefore, ___." · "Overall, the evidence supports the claim that ___."	therefore · overall · in conclusion

12 · Test-day game plan

Before the test

When	What to do
First: as soon as you can	**Get the real Part IV (light color) and Part V (temperature) numbers from classmates** and write them into your packet: your sheet is blank there (other teams did those parts), and a Part VII-style question needs them as evidence. Fix your worksheet slips with Guide 5 (Detected Errors).
3–4 days before	Read Guide 1 (Learning Guide), Parts 1–6; then reread Sections 3–6 of this guide.
2 days before	Level 1–2 mock questions in Guide 3 (Mock CER Questions) on a timer; score each with the Section 2 checklist.
1 day before	Level 3–4 mock questions; reread Sections 8 and 10 here.
The night before	Read Guide 4 (Cheat Sheet) twice, then the "Top 5 fixes" in Guide 5 (Detected Errors); say both equations aloud with the right energy labels. Then sleep.
The morning of	Write the 5 steps and both equations from memory (on scratch paper, if allowed).

During the test

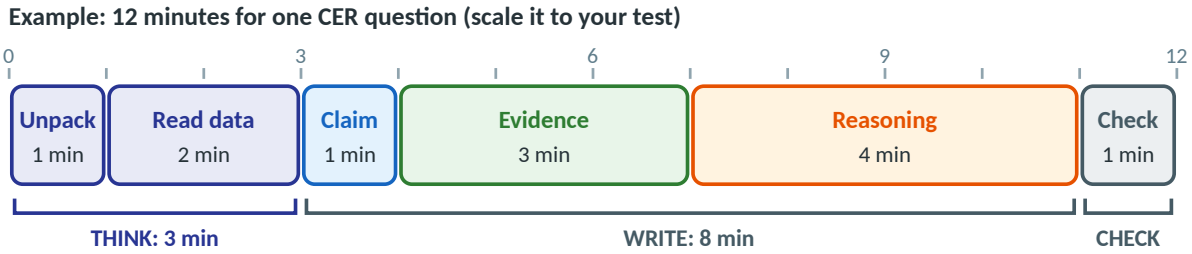


Figure 10. A time plan for one CER question. Reasoning gets the most time because it is worth the most points (3 of 6).

Unpack → data → C → E → R: circle the IV and DV, find the trend and 2–3 numbers, then write C, E and R, labeling each part [C] [E] [R] (or color-coding it). If the question says "photosynthesis and respiration", use both. Stuck? Write C and E first (3 points), then come back.

Final 30-second check

	Check	Points
☐	My claim is one sentence and answers the exact question (no "I think").	1
☐	I used at least 2 numbers, each with a unit and time basis — and the right oxygen number (DO difference vs O_2 production).	1
☐	I named the science principle(s): reactant/product, organelle, energy.	1
☐	I linked my evidence to my claim ("This shows… because… therefore…").	1
☐	I mentioned respiration happening at the same time (if O_2 or CO_2 is involved).	1
☐	I labeled [C], [E], [R] and did not claim beyond the data.	1
	Total	**6**

Quick self-check before the test

1. What does each part of a CER do?

Answer: **Claim** answers the question in one sentence; **Evidence** proves it with specific numbers and units; **Reasoning** explains WHY, using a science principle linked to the evidence.

2. At 100 % light, is the O_2 production 0.5976 or 0.4037 mg/L per hour?

Answer: 0.5976 is the O_2 production (gross); 0.4037 is the DO difference (net).

3. Why did the DO fall at 20 % light?

Answer: Respiration used more O_2 (0.1939) than photosynthesis made (0.0011 mg/L per hour).

4. Why does CO_2 at Mauna Loa peak in May?

Answer: It is a running total: respiration and decay add more CO_2 than photosynthesis removes until spring photosynthesis catches up.

You have got this. Your Part III data are solid and your conclusion was right. On test day, add the numbers and the WHY.